

Nagappa Siddgonde

Curriculum Vitae

Assistant Professor
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Personal Details

Name Nagappa
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Academic Qualifications

2016–2022 **PhD, in Aerospace Engineering**, *Indian Institute of Technology, Kharagpur.*
Thesis: Experimental and Numerical Characterization of Woven Fabric Composites Considering Void Defects. Advisor: Prof. Anup Ghosh
2013–2015 **Master of Technology, in Machine Design**, *Visvesvaraya Technological University, Belgaum.*
Thesis: Numerical Analysis of Low-Velocity Impact on Fibre Reinforced Laminated Composite Plates. Advisor: Prof. Shravankumar B Kerur
CGPA : 9.38/10
2010–2013 **Bachelor's, in Mechanical Engineering**, *Visvesvaraya Technological University, Belgaum.*
CGPA : 8.55/10
2007–2010 **Diploma, in Mechanical Engineering**, *Department of Technical Education Board, Bangalore.*
Percentage : 66.08 %

Research Interests

Area of research: Experimental mechanics, Multiscale modeling, Fracture mechanics, FE analysis, High temperature materials, carbon/carbon composites, Sustainable materials, Advanced Composite Materials, Multifunctional Composites, Textile composites, Ageing mechanism in thermoplastic composites.

Research Experience

July 2025- till **Assistant Professor at the Dept. Of Aerospace Engineering, IIT Madras, Chennai.**
June 2023 - **Research Associate at The University of Edinburgh, Edinburgh Scotland, UK.**
Nov 2024

Project: Advanced Composites Manufacturing

Sponsoring agency: Engineering and Physical Sciences Research Council (EPSRC)

Supervisor: Prof. Dipa Roy

Nature of work:

- Fabrication and testing of Advanced Composites using liquid resin infusion technique for marine and renewable energy applications.
- Fabrication and testing of E-Glass/Acrylic thermoplastic composites under both Dry and seawater ageing environmental conditions
- Coupon tests: Mechanical, thermal(DMA), and fracture tests.

March 2022- **Senior Research Fellow at SRIC, Indian Institute of Technology Kharagpur, India.**
March 2023

Project: *Fracture Control of CFRP Laminated Composites using Smart Materials Approach under Cryogenic Environmental Conditions.*

Sponsoring agency: Aeronautics Research & Development Board (AR&DB)

Supervisor: Prof. Anup Ghosh

Nature of work:

- An out-of-autoclave (oven) composite laminate fabrications lab has been set up for project work using carbon fiber prepreg material for aerospace applications.
- The material testing lab has been set up for composite material testing under extremely cold (cryogenic) environmental conditions (up to -80°C).
- Conducted Mode-I & Mode-II fracture test for CFRP composites under different loading and environment conditions.

2016-2022 **Graduate Research Assistant at Indian Institute of Technology Kharagpur, India.**

Supervisor: Prof. Anup Ghosh

- Developed a combined research framework for the experimental and numerical characterization of carbon/epoxy and E-glass/epoxy textile composites, with a focus on void defects.
- Developed various finite element-based 2D and 3D textile composite RVE models and investigate their mechanical, thermo-mechanical, and thermal properties
- Conducted various coupon tests to evaluate the mechanical properties of the 2D woven fabric composites.

Teaching Interests

Mechanics of Materials, Solid Mechanics, Aerospace Structures, Micromechanics, Mechanics of Composite Materials and Structure, Experimental Solid Mechanics

Teaching Experience

2025-till **Assistant Professor at Department of AE, IIT Madras.**

2016-2022 **Teaching Assistance at Department of AE, IIT Kharagpur.**

2013-2015 **Teaching Assistance under TEQIP-II at Basaveshwar Engineering College Bagalkot.**

Selected Skills

Programming Matlab, Python
CAE software Abaqus, Ansys, Digimat
CAD software Solid Edge, Auto-Cad, TexGen
Languages English, Kannada(native), Hindi

Publications

Journal Articles (SCI indexed journals)

- 2026 **Nagapp Siddgonde**, James A. Quinn, Conchur M. Ó Brádaigh, and Dipa Ray, Mode I Fracture Toughness and Compressive Properties of Glass Fibre Reinforced Acrylic Matrix Composites Before and After Seawater Ageing, *Polymer Composites (Accepted)*.
- 2024 **Nagappa Siddgonde**, Vikas Kaushik, and Anup Ghosh. Experimental and numerical characterization of e-glass/epoxy plain woven fabric composites containing void defects. *Aerospace Science and Technology*, volume 155, page 109731. Elsevier, 2024.
- 2022 Vikas Kaushik, **Nagappa Siddgonde**, Fumio Narita, and Anup Ghosh. Mode-i fracture control of unidirectional laminated composites using mfc actuators. *Journal of Intelligent Material Systems and Structures*, volume 33, page 2440–2453. SAGE, 2022.

- 2020 **Nagappa Siddgonde** and Anup Ghosh. Thermo-mechanical modeling of c/c 3d orthogonal and angle interlock woven fabric composites in high temperature environment. *Mechanics of Materials*, volume 148, page 103525. Elsevier, 2020.
- 2019 **Nagappa Siddgonde** and Anup Ghosh. Thermo-mechanical modeling of 3d woven fabric composites using two-step homogenization approach. In *ASME International Mechanical Engineering Congress and Exposition*, volume 59490, page V012T10A055. American Society of Mechanical Engineers, 2019.

Communicated Journal Article

- 2026 , L Gunaratne, **N Siddgonde**, J Maguire, D Morrisset, C E MacLeode, R M Hadden, D Ray, Improving Fire Performance and Thermomechanical Response of Glass Fibre/Acrylic Composites through Reactive Hybridisation with Polyphenylene Ether, (*submitted to Journal of Applied Polymer Science*).
- 2026 , R Wagmare, T Kuila, N C Murmu, **N Siddgonde**, M Doddamani, P Mallisetty, A review on Hydrogen storage (type IV) tank: Recent advancements and perspectives on fabrication, materials, treatments, and interfacial bonding strength between polymer liner, Aluminium boss, and CFRP composites, (*Submitted*).
- 2026 **Nagappa Siddgonde**, and Anup Ghosh, Experimental and numerical characterization of carbon/epoxy 5-harness satin composites considering the presence of voids using ANN and Monte-Carlo simulations, (*Under preparation*).

In Conference Presentations/proceeding

- 2024 **N. Siddgonde**, J. Quinn, A. Alapati, M.Devine, C. M. Ó Brádaigh and D. Ray, Investigation of Compressive and Interlaminar Fracture Properties of GF/Acrylic Thermoplastic Composites Under Sea Water Ageing Effect, *18th International Conference on Advances in Experimental Mechanics*, 3 – 5 September 2024 at University of Liverpool, UK.
- 2022 **N. Siddgonde**, and A. Ghosh, Two-step homogenization of 8-harness satin weave woven composite considering void defects, In *14th International Conference on Textile Composites (TexComp14)*, 14th -16th September 2022, at Kyoto Institute of Technology, Kyoto, Japan.
- 2019 **Nagappa Siddgonde** and Anup Ghosh. In *ASME International Mechanical Engineering Congress and Exposition*, volume 59490, page V012T10A055. American Society of Mechanical Engineers, 2019.
- 2019 **N. Siddgonde** and A. Ghosh. Thermo-mechanical modelling of 5-harness satin weave c/c composites at high temperature. In *9th ICMPC-2019*. 8th -10th March 2019 at Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad, 2019.
- 2015 Shravan B. Kerur, **Nagappa Siddgonde**, and Ravikant B. M. Numerical analysis of low velocity impact on woven fabric laminated composite plates. In *Proceedings of the IFERP International Conference:ISBN:9788192958066*, pages 16–20. IFERP at Bangalore, 2015.

Engagement With Professional Bodies

June 2014: The Institution of Engineers, IEI, India, (Lifetime Membership).

Reviewer: Journal of Composite Materials, MAM&S, ASME IMECE conference.

Fellowships & Awards

- 2016-2022 **Received MHRD Fellowship, from Government of India, as a PhD research scholar in Indian Institute of Technology Kharagpur..**
- 2019 **Received Institute Travel Grant for international conference, ASME International Mechanical Engineering Congress and Exposition 2019 held at Salt Lake City, USA.**

2013-2015 **Received Research Assistant Fellowship under TEQIP-II programme for two years during my M.Tech at Department of Mechanical Engineering, Basaveshwar Engineering College Bagalkot, Karnataka.**

The following skills gained from the past research work

Experimental Skills:

- **Mechanical tests:** *Tensile, compression, In-plane shear, Flexure, & ILSS*
- **Fracture tests:** *Mode-I & II Fracture tests*
- **Non-destructive Inspection:** *X-ray Microtomography, C-scan*
- **Thermal analysis:** *DMA, DSC*
- **Fractographic study:** *Scanning Electron Microscopic analysis*
- **Strain measurements:** *Digital image correlations, Extensometer, Strain gauges integrated with DAQ systems(P3 Vishay, System 7100, 8000).*
- **Physical property characterization:** *Acid digestion, Burn-off test and Density measurements*
- **Image processing:** *AxioVision, myVGL software, ImageJ, Inkscape*
- **Composite Fabrication Methods:** *VARTM, VARIM, CI tool, Prepreg autoclave system, Out of autoclave*

Computational Skills:

- Developed various finite element RVE models using Abaqus, Ansys and Texgen tool
- Developed Abaqus python scripts for pre- and post processing results analysis
- Implemented periodic boundary condition in Abaqus through Python codes
- Incorporated Void defects in textile composites
- Developed ANN model for woven fabric composites

Communication and Collaboration:

- Developed proficiency in preparing research manuscripts, funded project proposals, and laboratory reports.
- Presented research findings at several national and international conferences across multiple global venues.
- Served on the organizing committee of the international conference *ICTACEM-2017* at IIT Kharagpur.
- Delivered an invited lecture at the *Workshop on Polymer Matrix Composites – Design & Development (WPCD-25)*.
- Organized and coordinated technical activities, including invited lectures, research seminars, and Research Scholar Day events.

Laboratory Setup:

- Developed a composite fabrication & testing lab under the AR& DB project
- Contributed to the development and installation of a low-temperature environmental chamber (0 to -80°C) for a material testing setup used in fracture testing for aerospace materials.

Referees

Prof. Anup Ghosh

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IIT Kharagpur*

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Prof. Dipa Roy

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